

Multiple Choice (10 marks)

1. In a healthy person maintaining energy balance in equilibrium (i.e. mathematical equality of energy intake and energy expenditure) is generally achieved on a:
 - (a) 4-5 day basis
 - (b) 3-4 day basis
 - (c) 5-6 day basis
 - (d) Hourly basis
 - (e) Monthly basis
2. If links between various diseases are found, then future older adults will probably
 - (a) Have a shorter life expectancy
 - (b) Be less likely to seek medical advice
 - (c) Take fewer medications
 - (d) Pay far less for pharmaceuticals
 - (e) Have higher medical insurance premiums
3. Which of the following is a function of folate?
 - (a) Synthesis of fatty acids from glucose
 - (b) Conversion of oxaloacetate to phosphoenolpyruvate
 - (c) Carboxylation of pyruvate to oxaloacetate
 - (d) Conversion of fructose to glucose
 - (e) Synthesis of TMP (thymidine monophosphate)
4. A quasi species virus such as influenza and HIV has which of the following characteristics?
 - (a) Possesses only RNA
 - (b) A stable and unchanging genome
 - (c) Possesses only DNA
 - (d) Has a single, unsegmented genome
 - (e) Possesses RNA and DNA
5. The group that permits analysis of the entire natural history of HIV/AIDS is:
 - (a) Non-seroconverters
 - (b) Seroconverters

- (c) Short-term survivors with low levels of CD 4+ cells
- (d) Long-term survivors
- (e) Long-term survivors with high levels of CD 4+ cells

True / False (10 marks)

Answer True or False

- 6. True or False: Urea is the main nitrogenous compound found in human urine, accounting for the majority of its nitrogen content.
- 7. True or False: Water is recommended as a safe and effective option for cleaning the mouth.
- 8. True or False: Mutations causing achondroplasia are classified as recessive negative, impacting the function of normal alleles.
- 9. True or False: An oxygen consumption rate of 2 l/min in a male athlete indicates an energy expenditure of about 200 kJ/min.
- 10. True or False: Approximately 60% of grieving individuals are estimated to need professional assistance to cope with their loss.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the role of papillary muscles in preventing the eversion of the right atrioventricular valve, and analyze their importance in maintaining proper cardiac function.
- 12. Discuss the World Health Organization's recommendations regarding zinc supplementation, focusing on the specific populations that may benefit, particularly men with high physical activity levels.

End of Paper